IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

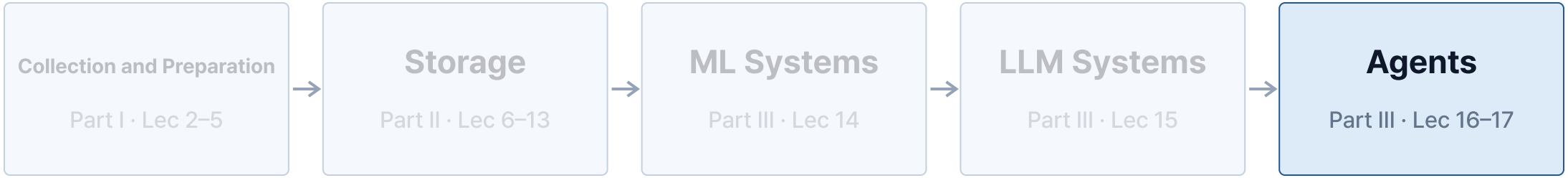
17.

Data Agents

Sangwoo Mo · POSTECH

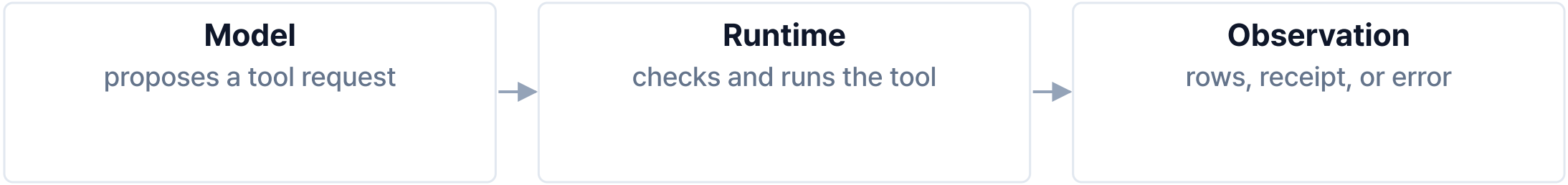
Industrial & Management Engineering

Where We Are



Part III connects stored data to predictions, answers, and actions

Recap: A Tool Request and Its Execution



The database result becomes the observation for the next decision

Outline: Data Agents

01

Understanding the Question

define the metric using the production records

02

Checking and Running a Read

check the query and answer from returned values

03

Making an Approved Change

review one change and verify its stored result

01 · Understanding the Question

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The Factory Request

A data agent uses tools to read records and make permitted changes

P-17
machine_id = 1

→

Read
yield over recorded runs

→

Possible change
if yield is below 85%
prepare an inspection request

The read supplies evidence for a proposed change

What Does “Yield” Mean?

Here, yield means good units divided by all produced units

Good units
items that passed inspection

→

Total units
all produced items

→

Scope
both recorded P-17 runs

The metric and selected records determine the answer

The Production Records

P-17 has two runs with different output sizes

P-17 = machine_id 1

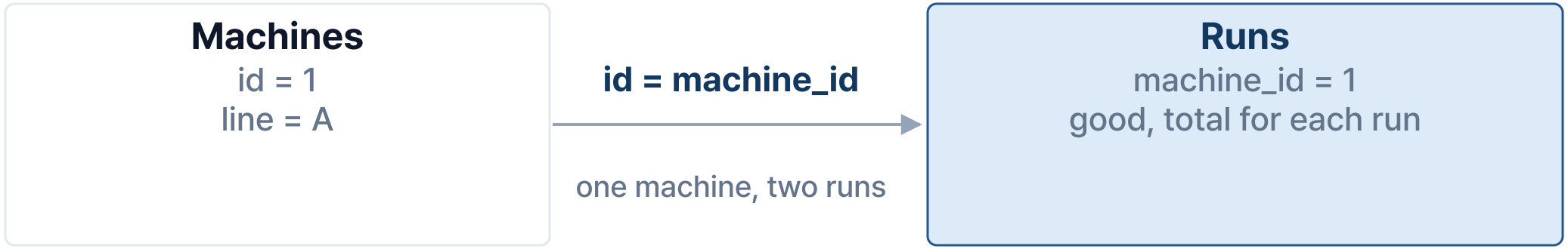
run	machine_id	day	good	total
1	1	2026-09-01	80	100
2	1	2026-09-02	9	10

One row records one production run

The second run produces one tenth as many units as the first

Tables and Machine IDs

The machine ID connects a machine to its production records

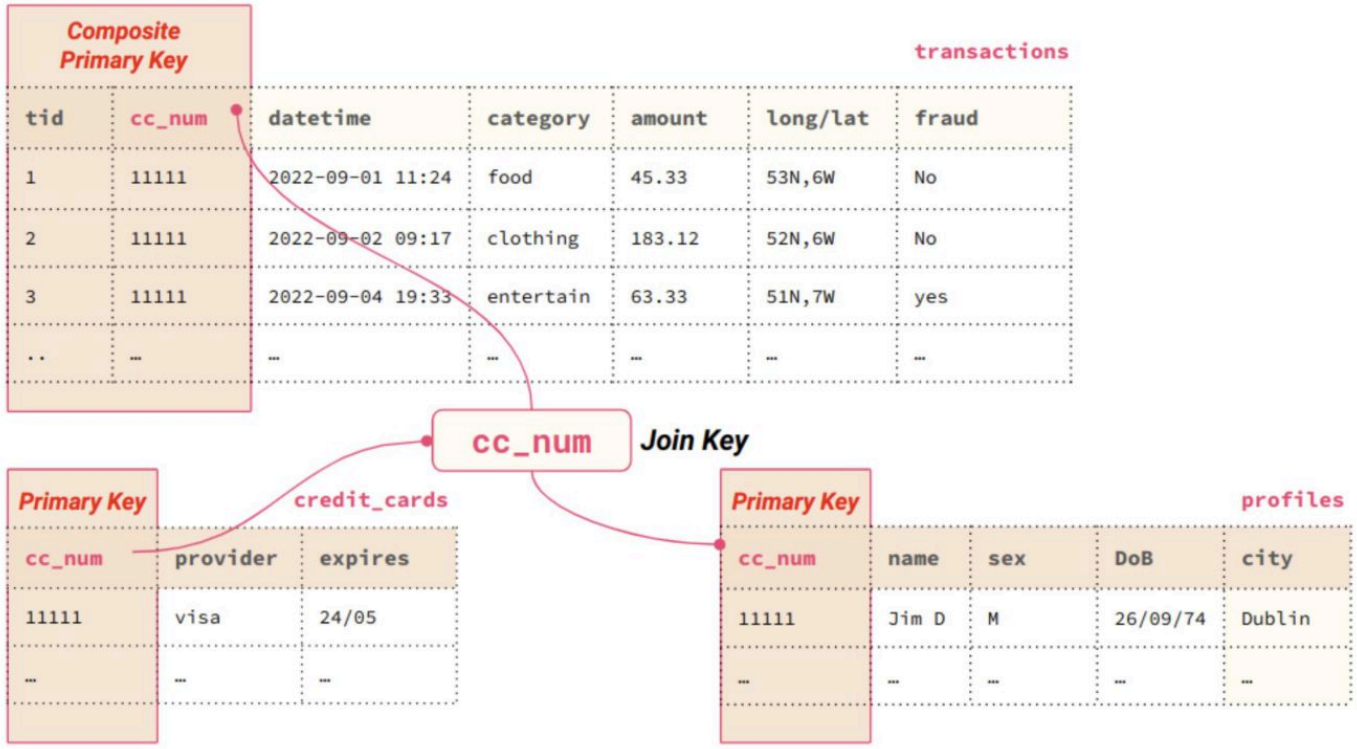


“P-17” is the case label for database ID 1

The query selects the two Runs rows for machine_id 1

Keys and Join Relationships

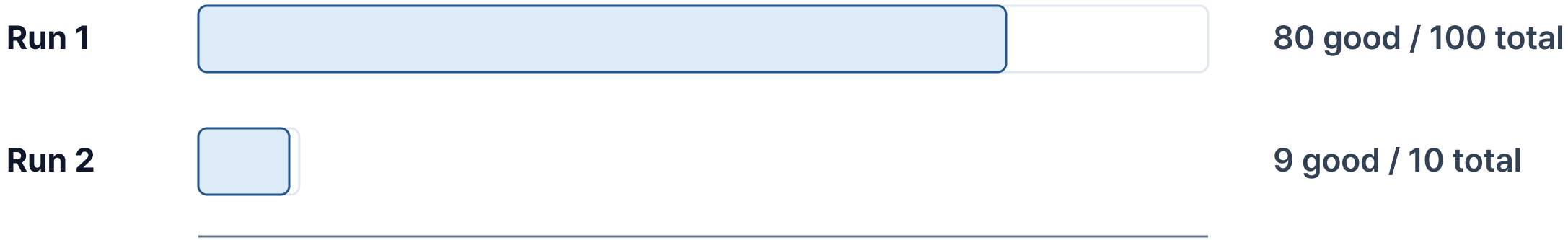
Table relationships determine which records a query will combine



The join key connects records, while the primary key identifies a row

Yield by Hand

Add the good counts and total counts, then divide

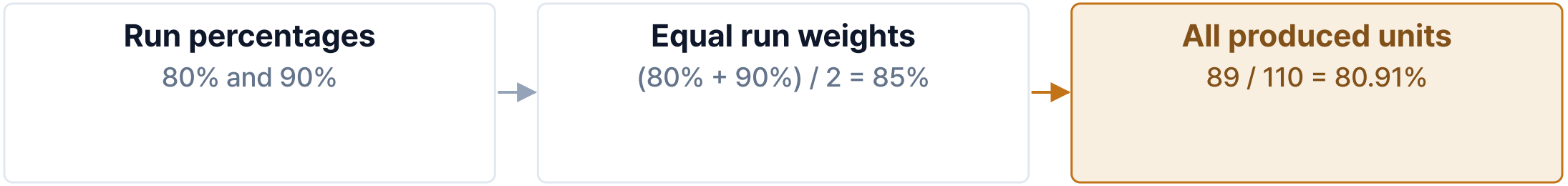


(80 + 9) / (100 + 10) = 89 / 110 = 80.91%

89 good units out of 110 total units gives 80.91% yield

Caveat: Averaging Run Percentages

Giving the two runs equal weight answers a different question



The larger run must contribute more to the combined yield

Join Multiplicity Changes Totals

A one-to-many join can duplicate one run before the sums are calculated

Original runs	Run 1 has two tags	Joined totals
89 good / 110 total	80/100, 80/100, 9/10	169 / 210 = 80.48%

```
SUM after the join: (80 + 80 + 9) / (100 + 100 + 10)
```

Joining tags first gives the first run twice its intended weight

Evidence for a Query Check

Keep one numerical counterexample for each distinct failure

Condition	Returned yield	What went wrong
Correct machine, raw runs	80.91%	89 / 110
Equal weight per run	85.00%	Different metric
Run 1 duplicated by join	80.48%	Different row weights

SQL can run successfully while answering a different metric or weighting the wrong rows

Text-to-SQL

The model proposes a query using the question and table definitions

Question
P-17 yield over both runs

→

Table definitions
Runs: machine_id, good, total

→

SQL proposal
filter ID 1, then sum counts

The SQL proposal still needs checking before execution

The SQL Proposal

One filter selects P-17, and two sums calculate its yield

```
SELECT SUM(good) AS good, SUM(total) AS total,
       ROUND(100.0 * SUM(good) / SUM(total), 2) AS
yield_pct
FROM Runs
WHERE machine_id = 1;
```

good	total	yield_pct
89	110	80.91

The database adds the counts for machine 1 before dividing

02 · Checking and Running a Read

01

Understanding the Question

define the metric using the production records

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Checking and Running a Read

check the query and answer from returned values

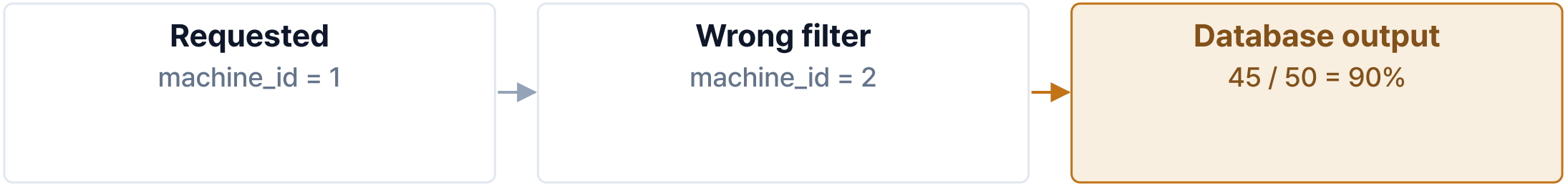
03

Making an Approved Change

review one change and verify its stored result

Why Check the Proposal?

A valid query can still select the wrong machine



Successful execution does not confirm that the request was understood

Checks Before a Read

Match the request and use only permitted data

Meaning
machine 1, both runs
SUM(good) / SUM(total)

→

Permission
caller may read these rows

→

Execution limit
bounded time and result size

Meaning and access require checks beyond SQL syntax

Read and Write Permissions

Permission to inspect production does not grant permission to create repairs

Read	Write
Analyst read production counts no repair creation	**Line A operator** create a repair for machine 1 cannot create one for line B

The database or tool server enforces the caller’s permissions

The same user can have different permissions for reads and writes

Read-Only and Row Scope

Preventing writes does not decide which machines a caller may read

Read-only connection
SELECT is allowed

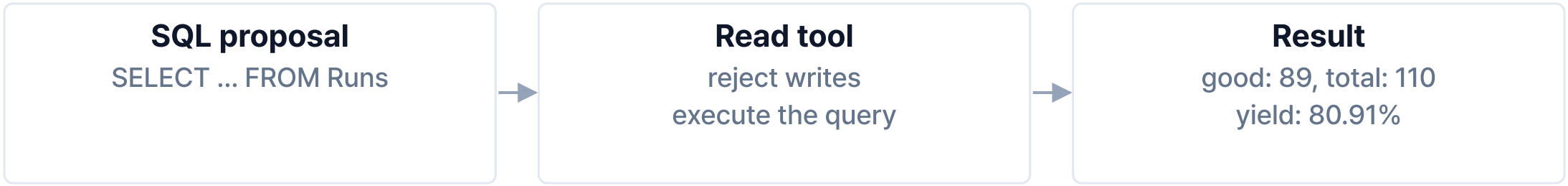
Caller scope
line A machines only

Permitted tool
server enforces line A

A read-only query can still return a line B row unless row access is enforced

The Read Tool

The runtime sends the checked query to a read-only database operation



The tool returns database rows rather than an invented number

Demo: Read P-17 Yield

Supplied SQL, no model inference; SQLite executes the read

1. Run the supplied query. Compare its counts with 89 / 110.
2. Change machine_id to 2. Explain why the new answer does not answer the request.

```
Type code or click Starter
```

```
Run code to inspect its output.
```

Run | Starter | Reset

Ready

LangChain.js · JavaScript · Ctrl/Cmd+Enter · Real library execution with local data

Answer: Read P-17 Yield

The stored counts agree with the hand calculation

Selected machine	good	total	yield_pct
1 (P-17)	89	110	80.91
2 (different machine)	45	50	90.00

Only machine 1 answers the original request

A Failed Read

Use the observed error to correct the query or stop

Proposal
SELECT yield FROM Runs

→

Database error
no such column: yield

→

Correction
calculate from good and total

The answer needs a successful, permitted database read

No Rows and a Zero Total

An unavailable ratio must not be reported as zero yield

Selected records	SUM(good)	SUM(total)	Yield
No matching runs	NULL	NULL	NULL
A run with no production	0	0	NULL
P-17 original runs	89	110	80.91%

No observed production is different from observed production with no good units

The Evidence for the Answer

Report the values and scope used in the calculation

Source
machine 1
both recorded runs

→

Calculation
89 / 110 × 100

→

Answer
80.91%, below 85%

The answer stays tied to the selected records

A Consistent Read

Calculate related counts from the same database snapshot

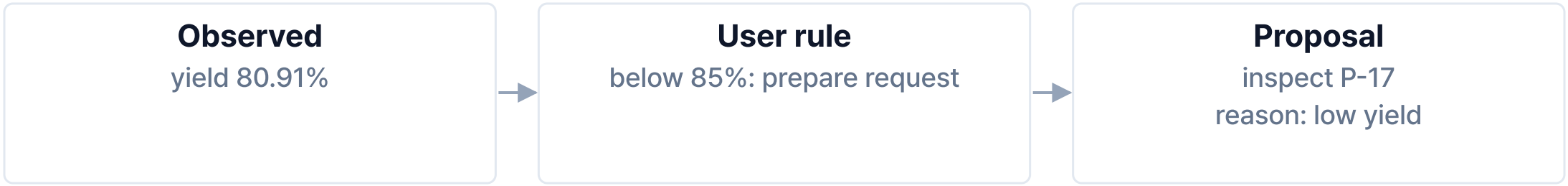
Original snapshot	Concurrent new run	Mixed observations
good 89 / total 110	+10 good / +10 total	89 / 120 = 74.17%

```
Before: 89 / 110 = 80.91%; after: 99 / 120 = 82.50%
```

One query returns 89/110 before the insert or 99/120 after it, rather than 89/120

A Low Yield Is Evidence for Inspection

The threshold triggers a proposed request, not a diagnosis



The requested change follows the user's rule and observed result

03 · Making an Approved Change

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check the query and answer from returned values

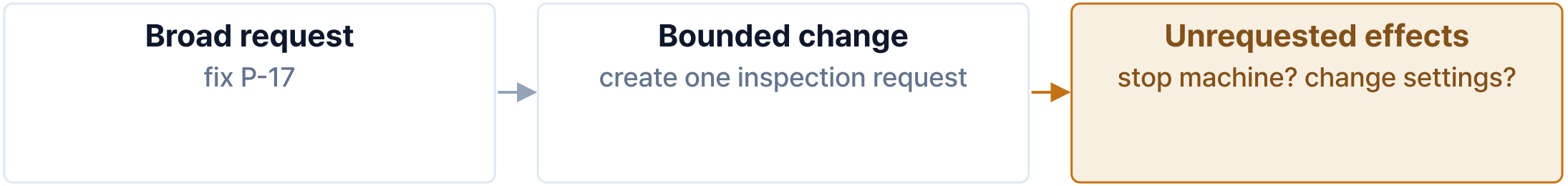
03

Making an Approved Change

review one change and verify its stored result

Why Review the Change?

"Fix P-17" does not identify what the system may change



The proposed effect must be concrete before approval

The Proposed Repair Request

The preview identifies the target and values to be stored

One new row in Repairs

request_id	machine_id	reason	status
req_42	1	low yield	open

A repair request records work to do. It does not stop or alter the machine.

The preview makes one database change reviewable

Design: Approval Before Writing

The application waits for a decision on this exact change



The approval gate is a design step, separate from the database demo

Approval and Permission

An approved request must still fit the caller's allowed scope

Approved proposal
create a repair for machine 1

→

Tool permission
line A machines only

→

Decision
machine 1: allowed
machine 3, line B: reject

Approval does not expand database permissions

The Exact Approved Change

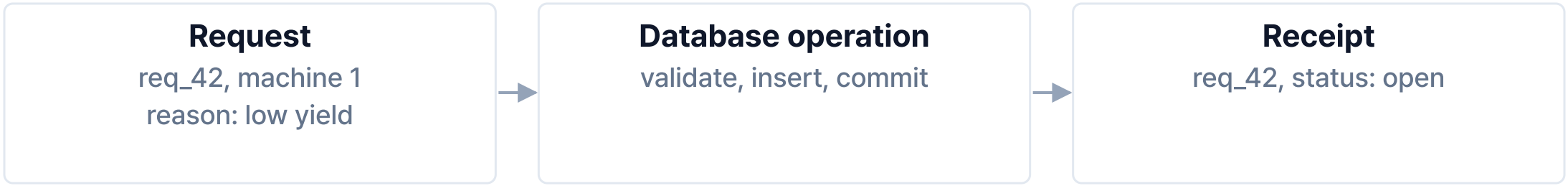
Approval belongs to the reviewed target and payload, not to any later tool call

Reviewed request	Actual request	Decision
req_42, machine 1, low yield	Same ID and same values	Execute within permission
req_42, machine 1, low yield	machine 3 instead	Reject changed target
req_42, machine 1, low yield	reason changed	Reject changed payload

Changing the target or reason creates a different action from the one reviewed

The Write Tool

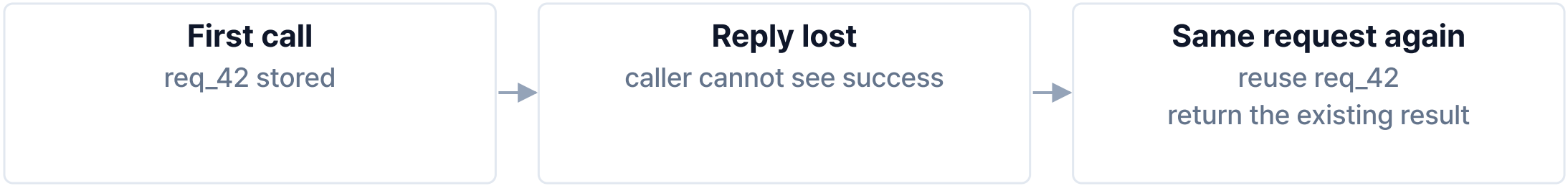
Create one repair and return its stored result



A successful receipt must correspond to a stored row

Caveat: Repeating a Request

A missing reply must not lead to a second repair request



The repeated request should refer to the same stored change

Demo: Create One Repair Request

Local SQLite write only; this demo has no approval gate

1. Run two calls with req_42. Count the stored repair rows.
2. Change machine_id to 3 (line B). Observe the permission error.

```
Type code or click Starter
```

```
Run code to inspect its output.
```

Run | Starter | Reset

Ready

LangChain.js · JavaScript · Ctrl/Cmd+Enter · Real library execution with local data

Answer: Create One Repair Request

Two calls refer to one stored row

Call	Created a row?	Stored rows
First req_42	yes	1
Same req_42 again	no	1
Machine 3 (line B)	permission error	0 in a fresh run

The repeated call returns the existing request

The Stored Request

Verify the row before reporting that the change succeeded

One new row in Repairs

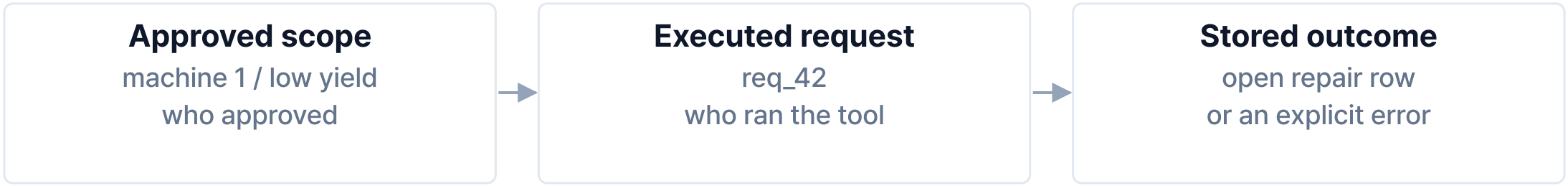
request_id	machine_id	reason	status
req_42	1	low yield	open

A repair request records work to do. It does not stop or alter the machine.

The completion message follows the database record

The Action Record

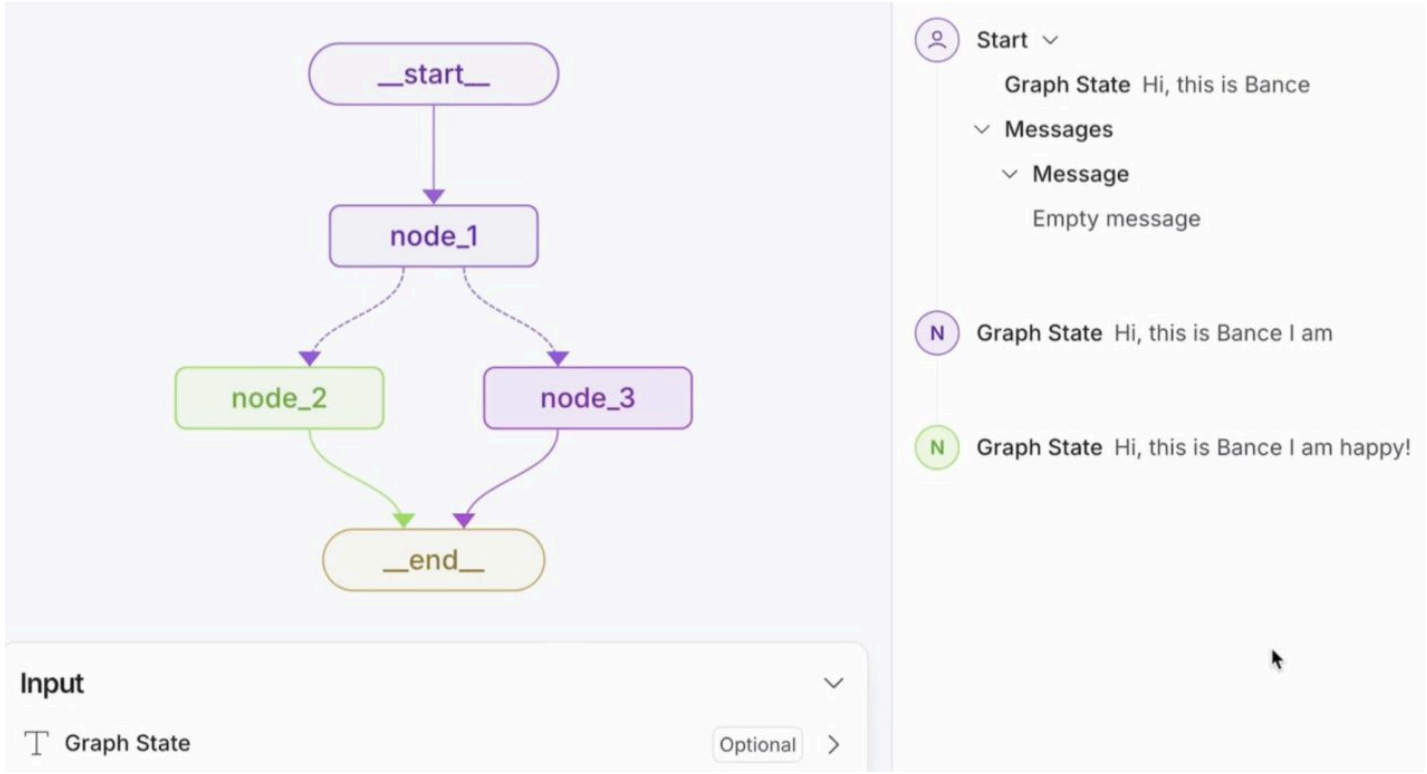
Keep the decision and outcome linked to the request



The record connects the intended change with its verified result

From Run View to an Audit Record

A path and message view helps locate a request; the database confirms its stored effect



An execution trace is useful evidence, but the stored row confirms the database change

Takeaway: Meaning, permission, and a verified result

1. Check metric, join multiplicity, empty data and a consistent snapshot
2. Enforce row permissions outside model instructions and answer from observed rows
3. Approve the exact payload, preserve request identity and verify the stored effect